CADEN BURKHARDT

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Education

B.S. in Honors Astronomy & Astrophysics and Interdisciplinary Physics, Minor in Climate & Space

University of Michigan, Ann Arbor, MI, Expected Graduation May 2026

- **GPA: 3.82/4.0**, Major GPA: 3.92/4.0
- **Honors/Awards:** University Honors all semesters, James B. Angell Scholar, Sophomore Honors Award

Presentations & Publications

Presentations

- **Burkhardt, C.**, Oey, M. S., Han, F., & Deman, J. (2024). Gaia Proper Motion Velocities of Wolf-Rayet Stars in the Large Magellanic Cloud. Bulletin of the AAS, 56(7).

Publications

- **Burkhardt, C.**, Oey, Han, F., Oey, M.S, et al. (2025). Kinematics of Wolf-Rayet Stars in the LMC: Clues to Subtype Origins. *Manuscript submitted for publication.*
- Han, F., Oey, M. S., & **Burkhardt, C.** (2025). Gaia Proper Motion Velocities of WC and WN3/O3 Stars in the LMC. *Manuscript submitted for publication.*
- Monnier J. D., Cutler, J., Meyer, M., **Burkhardt, C.**, Dubey., A, et al. "STARI: STarlight Acquisition and Reflection toward Interferometry," *Proceedings of the AIAA/USU Conference on Small Satellites*, Advanced Technologies 2 - Research & Academia, SSC25-RAIII-08. https://digitalcommons.usu.edu/smallsat/2025/RA-S8-2025/5/.

Technical/Research Experience

Undergraduate Research Assistant with Professor Sally Oey **November 2022 - Present**

Researching Wolf-Rayet (WR) stars in the Large Magellanic Cloud (LMC), using their kinematics to discover more about massive star evolution.

- Developing Python scripts to query data from Gaia DR3 catalog and crossmatching it with a Wolf-Rayet (WR) star catalog to calculate proper motion velocities for WR stars in the Large Magellanic Cloud.
- Analyzing velocity distributions and utilizing statistical tests to compare the velocities of different WR subtype populations to better understand WR stellar evolution.
- Wrote results into two papers (*Burkhardt et al.,2025 & Han et al., 2025).*

STarlight Acquisition and Reflection toward Interferometry (STARI) **March 2025 - Present**

Developing the optical payload for STARI, the first space mission led by University of Michigan Department of Astronomy, to demonstrate the technological viability of space-borne optical interferometry.

Laboratory Assistant with Professor John Monnier

- Designing and fabricating prototype components using both 3D printing and machining on a manual milling machine
- Writing code using Motor Control Language (MCL) to control picomotors in conjunction with imaging processing to analyze images and perform closed loop control of optical mirrors.
- Performing optical breadboarding & laser alignment to test prototype optical designs

Michigan Mars Rover Team (MRover) **September 2022 - Present**

Providing student leadership for the largest interdisciplinary robotics team at the University of Michigan, which competes yearly with the top teams from across the globe in two international competitions.

Vice President *May 2025 - Present*

- Leading over 150 members, focusing on member recruitment and community building initiatives.
- Directing the implementation of the team's Organizational Development Plan.
- Organizing new member training curriculum with subsystem and branch leads.

Senior Advising Mentor *May 2024 - May 2025*

- Provided leadership mentoring, design feedback, and technical advice for subsystem leads and mechanical members.
- Trip leader at the Canadian International Rover Competition (CIRC) – Team placed 2nd.

Instrumentation & Sample Handling (ISH) Subsystem Lead *May 2023 - May 2024*

- Led subteam in the design and fabrication of ISH subsystem for the University Rover Competition at the Mars Desert Research Station (Hanksville, Utah).
- Spearheaded subsystem redesign, incorporating a new-on-rover life detection assays and RAMAN spectrometer, improving the rover's ability to analyze soil samples for signs of life.

Instrumentation & Sample Handling Subsystem Member *September 2022 - May 2023*

- Employed Siemens NX CAD software to design and iterate on multiple system components, including a sealable sample cache and camera mounts.
- Conducted comprehensive testing on all system components, including cameras, spectral sensors, UV LEDs, and heating elements to ensure smooth system performance.

Additional Leadership Experience

Student Astronomical Society (SAS) **August 2022 - Present**

President *April 2025 - Present*

- Leading weekly club meetings, organizing and preparing engaging presentations and speakers while fostering a welcoming environment.
- Coordinating with Astronomy Department to organize professional development opportunities for undergraduate students
- Running public outreach events including bi-weekly open houses which utilize the Angell Hall planetarium and 0.4m telescope.

Secretary *April 2024 - April 2025*

- Planned and coordinated club trips to various locations such as local observatories, a Dark Sky Preserve, and Astronomy at the Beach (a regional amateur astronomy event).
- Managed club records, including writing weekly announcements, managing the email list, tracking attendance, and taking club meeting minutes.

Undergraduate Research Symposium (URS) **August 2025 - Present**

Internal Committee Member *August 2025 - Present*

- Raising over $5,000 to fund ten travel grants to allow for students with outstanding undergraduate research to present at external conferences.
- Organizing research fundamentals workshops to teach essential research skills to undergraduate scientists.

Michigan Magic Club **September 2023 - Present**

President *September 2024 - Present*

- Organizing performances within the local community including at a Mitchell Elementary School PTO event.
- Developing lesson plans to teach interested students magic tricks and sleight of hand techniques.

Skills

Programming/Software

- Programming Languages: Python (Pandas, Matplotlib, Scikit-learn, Numpy, Astropy), ADQL, UNIX Shell Commands, Arduino
- CAD: Siemens NX, Solidworks, Autodesk Inventor
- Other Software: Ansyst STK (Systems Tool Kit), SAOImage DS9, MESA - Web

Prototyping/Hardware

- Trainings/Certifications: Wet/Chemical Laboratory User Certification, Laser Safety Basic Training
- Machining: General Workshop Certification (*Bandsaw, Drill Press, etc*), Manual Mill, 3D printing
- Misc: Operator on 0.4 m Ritchey-Chretien reflector telescope